Fiscal Year:	FY 2022	**Task Last Updated:**	FY 08/22/2022
PI Name:	Rojas-Pierce, Marcela Ph.D.		
Project Title:	Membrane Contacts in Plant Gravity Perception		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27695-7612	**Congressional District:**	4
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	10/29/2018	**End Date:**	10/28/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 10/21/2022 per NSSC information (Ed., 8/24/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0145		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants grown during spaceflight experience conditions of microgravity, ionizing radiation, and reduced gas exchange. If we are to use plants in advanced life-support systems during long space missions, it is imperative that we understand how plants respond to these varying conditions to ensure their productivity. Plant growth on Earth is shaped in great part by the force of gravity, while conditions of microgravity experienced in space result in dramatic changes in organ orientation and growth. The plant response to gravity, or gravitropism, is also important for the growth angle of root and shoot branches, and therefore, it also impinges in crop architecture and productivity on Earth. The effects of microgravity on plant growth at the cellular level are not yet fully characterized, and the molecular mechanisms by which plants perceive gravity are still unknown. A wealth of state-of-the-art genetic, imaging, and omics tools in the model plant Arabidopsis thaliana are ripe to provide the clues for this elusive system in plant biology, and novel players have recently been identified. We propose here to characterize the role of two protein families predicted to mediate contacts between the endoplasmic reticulum (ER) and other organelles as candidates for important roles in gravitropism. These proteins were selected based on our previous NASA-funded research where we characterized the role of vacuoles in gravitropism. This work led us to propose the hypothesis that membrane contacts between amyloplasts and vacuoles and/or ER have a role in gravity sensing in shoot statocytes. We plan to test this hypothesis directly by addressing the specific aims to 1) determine the function of candidate membrane tethers in membrane contacts and gravity sensing; and 2) determine the effects of spaceflight on vacuole fusion and ER morphology. A reverse genetic approach combined to fluorescence microscopy and mutant characterization will be used to characterize the role of selected tethers. The flight experiment will use well-established methods for Arabidopsis growth in Biological Research in Canister-Petri Dish Fixation Unit (BRIC-PDFU) units that are suitable for chemical treatment and fixation. Plant sub-cellular phenotypes will be analyzed after sample return to Earth. Our ground-based research will address the most important knowledge gap in gravitropism at the cellular level and in specific cell types of the Arabidopsis shoot. In addition, the proposed flight experiment will result in multiple outcomes beyond documenting the effects of spaceflight on organelle fusion. These include probing the use of chemical inhibitors for plant cell biology experimentation in space and testing recently described tissue clearing protocols for visualization of cellular organelles in space-grown plants. Our research will train young scientists in space biology research including undergraduate students, and therefore contribute to the development of human resources in STEM (Science, Technology, Engineering, and Mathematics).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will contribute to the understanding of the molecular mechanisms that control the response of plants to gravity. The identification of candidate proteins involved in membrane contacts and gravitropism will enhance the fundamental understanding of how plants perceive and respond to gravity signals.
Task Progress:	This last year we completed the analysis of the flight experiment at the International Space Station (ISS). Progress for Aim 1) Determine the function of new candidates in gravity sensing. This past year we have focused on two genes with putative roles in membrane contacts. We generated functional green fluorescent protein (GFP)-tagged lines that show patterns of localization consistent with a role in organelle-organelle contacts. Phenotypes of single and double mutants support the hypothesis of these proteins being important for gravity sensing in plants. Progress for Aim 2) Determine the effects of spaceflight on vacuole fusion and endoplasmic reticulum (ER) morphology. This past year we analyzed results from out flight experiment at the ISS. The flight experiment in July 2021 involved the treatments of two genotypes in the ISS with a Wortmannin solution. Samples were fixed in a paraformaldehyde solution, stored at 4C and subsequently returned to Earth. A manuscript describing these results has been accepted for publication in the journal, Microgravity. Careful analysis of the vacuole number in root cells of treated plants indicated that the spaceflight environment had mild effect in vacuole fusion and differences between flight and ground control were only detected in epidermal cells of the root transition zone. Our research successfully demonstrated the feasibility of space experiments involving chemical treatments of Arabidopsis seedlings, tissue fixation, and visualization of GFP-tagged markers.
Bibliography Type:	Description: (Last Updated: 11/17/2022)
Articles in Peer-reviewed Journals	Wang M, Danz K, Ly V, Rojas-Pierce M. "Microgravity enhances the phenotype of Arabidopsis zigzag-1 and reduces the Wortmannin-induced vacuole fusion in root cells." npjMicrogravity. 2022 Sep 6;8(38). https://doi.org/10.1038/s41526-022-00226-3 , Sep-2022